Incorporating Health Outcomes into Building Decarbonization Plans

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Aim

This project aims to **quantify** the impacts of active **ventilation** on **health outcomes**.

★ Develop a Python-based framework that measures health impact.

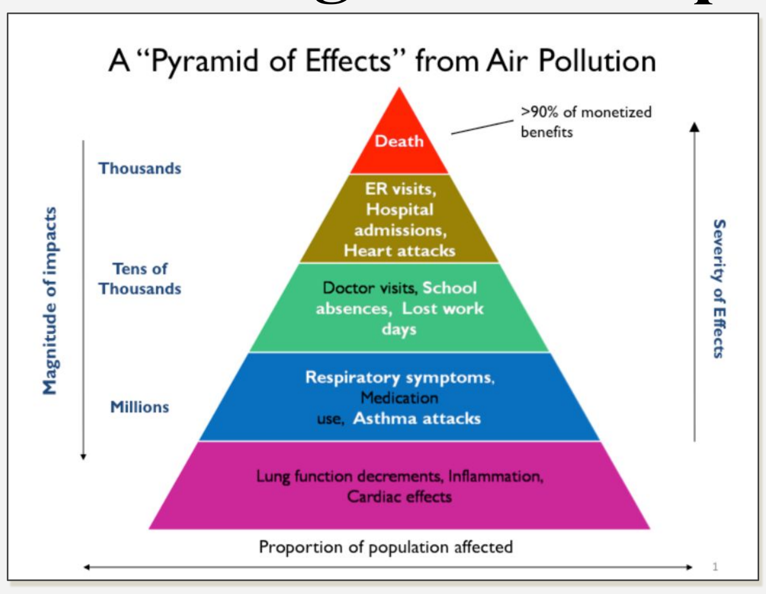
Background & Motivation

While **Building decarbonization** pathways are often designed with efforts to **reduce** greenhouse gas (GHG) **emission** goals in mind, they also have tangible **socioeconomic impacts.**

★ Heating, Ventilation, and Air Conditioning (HVAC) retrofits affect levels of indoor **pollutants** like **fine particulate matter ($PM_{2.5}$)**.

★ Renters may bear the hidden **costs of indoor air quality changes.**

Calculating Health Impacts: Asthma, ER Visits, & Mortality



EPA health impacts associated with poor air quality.

★ The Environmental Protection Agency (EPA) identifies **health metrics**.

★ The Log-Linear Concentration Response Model uses the **relative risk** of each health outcome to track changes in health impact across changes in $PM_{2.5}$ concentrations.

★ We can quantify these outcomes through the **cost of illness**. According to the Central Estimate of Value Per Statistical Incidence.

- Mortality: $8,705,114
- ER Visit: $600-$1,200
- Asthma Onset: $182,681

$$\frac{dC_{\text{in}}}{dt} = \frac{1}{V}\Big[S_{\text{indoor}} + Q_{\text{inf}} \cdot P_{\text{inf}} \cdot C_{\text{out}} + Q_{\text{vent}} \cdot (1-\eta) \cdot C_{\text{out}} + k_{\text{resus}} \cdot V \cdot C_{\text{in}} - (Q_{\text{inf}} + Q_{\text{vent}}) \cdot C_{\text{in}} - Q_{\text{recirc}} \cdot \eta \cdot C_{\text{in}} - k_{\text{dep}} \cdot V \cdot C_{\text{in}}\Big]$$

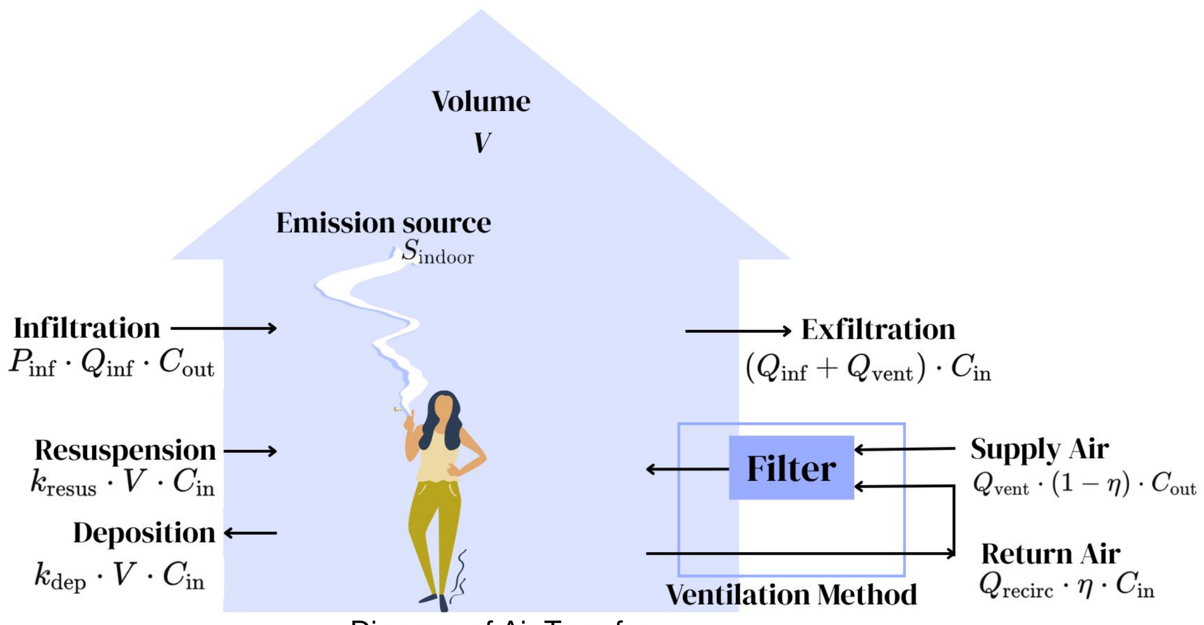


Diagram of Air Transfer

Developing an Air Quality Model

Building from the general air mass balance or **$PM_{2.5}$ mass transfer equation**, we incorporate the use of a filter, as well as an indoor emission sources with unique emission rates.

★ Minimum Efficiency Reporting Value (MERV) ratings increase according to their efficiency to filter fine particulate matter.

- **MERV 8:** 85% of $PM_{2.5}$ enters
- **MERV 13:** 15% of $PM_{2.5}$ enters

★ Outdoor air data for the past decade was sourced from the EPA's Air Data - Daily Air Quality Tracker.

★ Indoor sources were used to simulate lifestyle "scenarios".

Scenarios

Daily activities emit PM2.5 into indoor air. An average resident, a non-smoker, cooks 37 minutes a day and cleans for 6 hours a week.

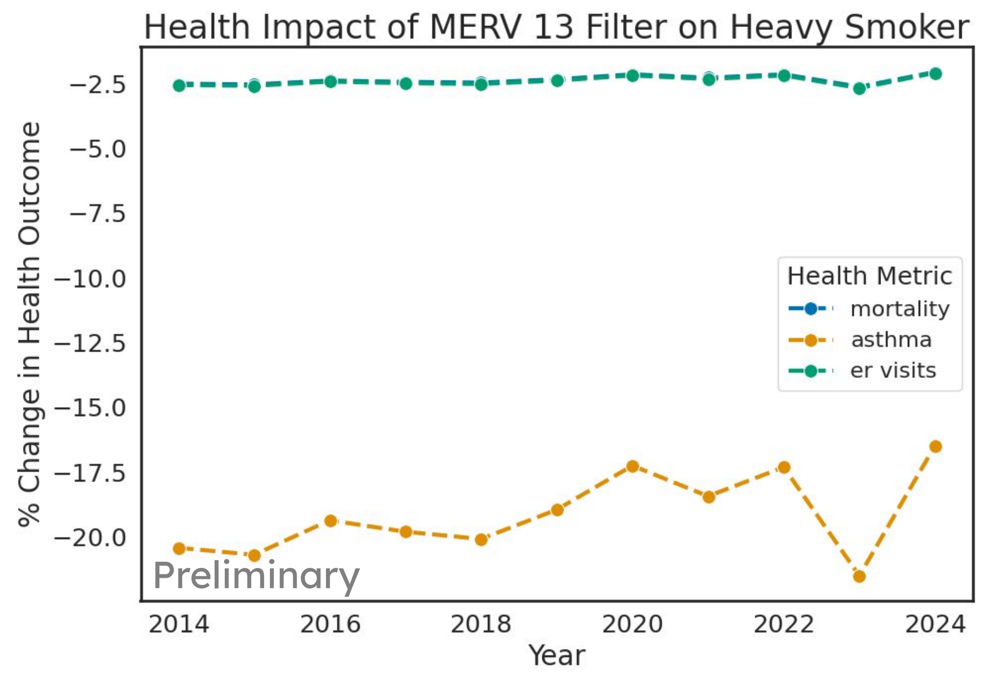
★ Heavy Smoker
- 2 hours/day smoking

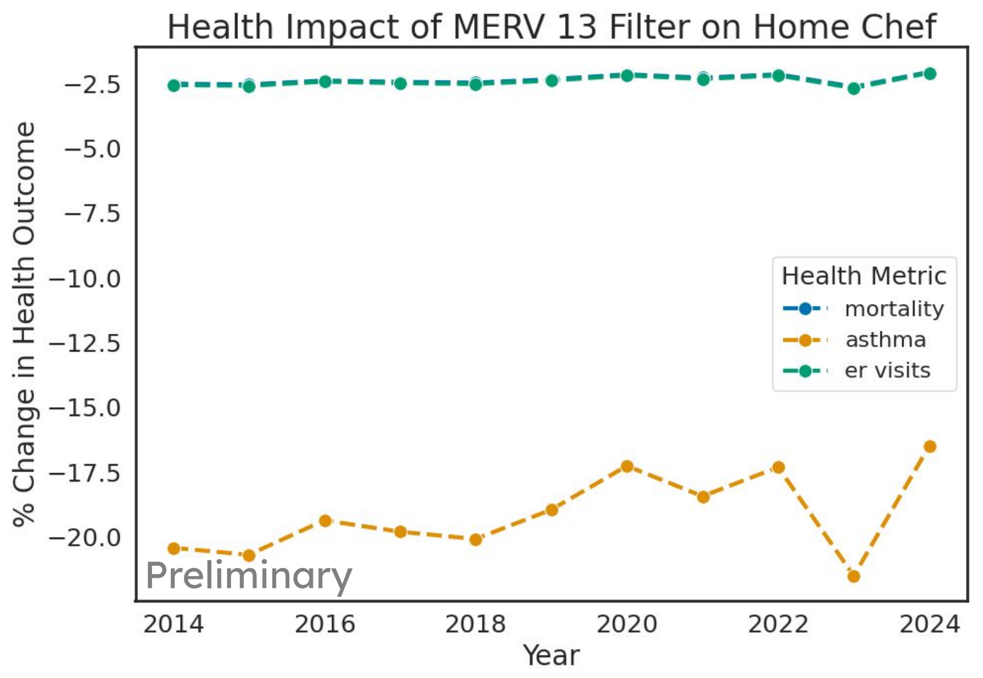
★ Excessive Cleaner
- 5 hours/day cleaning

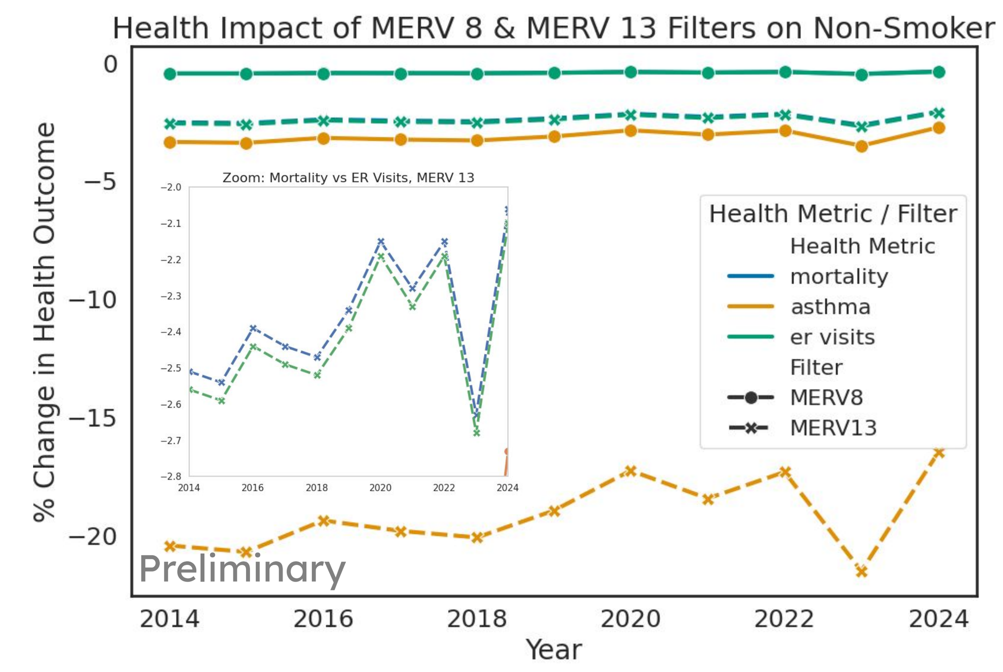
★ Home Chef
- 5 hours/day cooking

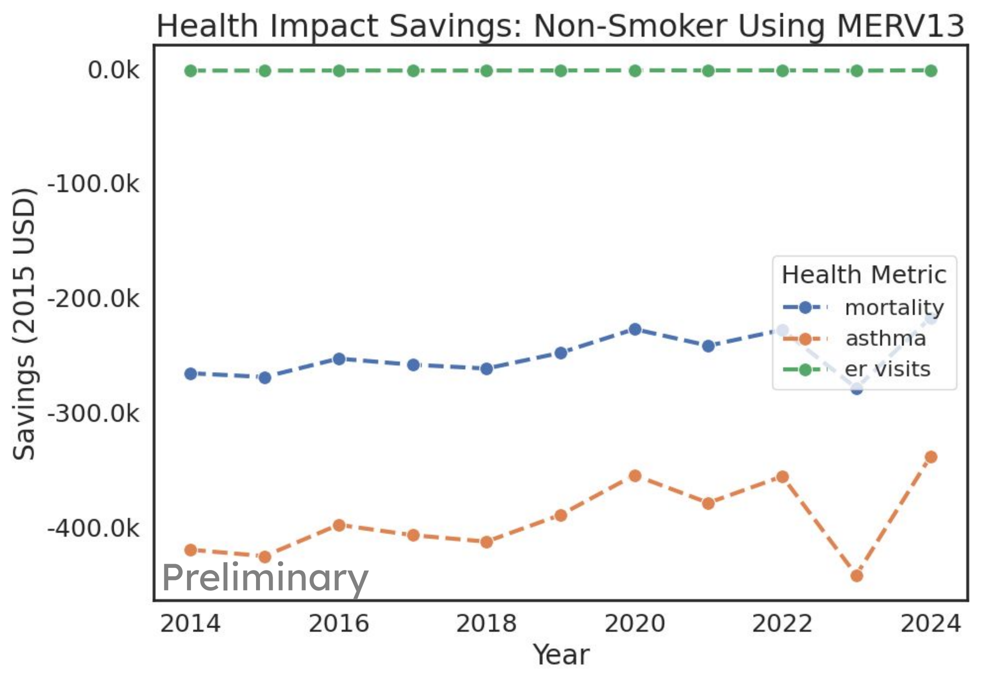
Comparing Health Impact Across Scenarios

Simulating a 10 story residential building with 74 units and about 300 occupants, we tracked the health impacts of our three scenarios. As seen in the top two plots, the lifestyle changes were outweighed by the dominating impact of air filtration.









ER visits and Mortality had very similar relative risks, so their change in health impact is incredibly similar, as seen in the MERV 8 & MERV 13 comparison plot (lower left).

Cost of Illness

Case Study

★ For a building of nearly **300 occupants**, the susceptible population, children aged 0 to 17, is estimated to be half the occupancy, 150.

★ The cost of illness metric shows that the **20% decrease in Asthma onset** associated with a **MERV 13** filter (lower left plot) could reduce the populations' lifetime Asthma-related costs by **~$400k**.

Next Steps

★ Incorporate into a study identifying high-impact retrofits within a simulated West Harlem block with eight buildings

★ Compare the health impact when using different HVAC and Ventilation strategies.

References

US Environmental Protection Agency (EPA). (2024b, April). Environmental Benefits Mapping and Analysis Program User's Manual. https://www.epa.gov/system/files/documents/2024-05/benmap_user_manual_v_0.5_508.pdf

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